

① Experiment 1 → Study of LAN Topology, Network devices and Address configuration

Objective: →

- 1) To study the LAN topology used in college Campus.
- 2) To identify networking device & understand their function.
- 3) To analyze wired & wireless Communication method used in the network.
- 4) To measure speed of Network for wired & wireless connections.
- 5) To obtain MAC, IP Address & Subnet mask.

→ Problem Statement - 1

- Draw & identify LAN topology of college Campus & clearly label all components such as router, switches & etc.

→ Solⁿ → The College Campus network observed this study consists of:

- * Layer-3 Core switches in the Server Room.
- * Managed by PoE+ Switches.
- * Wireless Access point (APs)
- * Ethernet port in Classroom & Laboratories.

① → Network Topology ②

Hybrid Hierarchical Topology which Combines:

- 1) Tree Topology.
- 2) Extended Star Topology.
- 3) Ring Topology (server room).

→ The network hierarchical layer

- ① Core layer.
- ② Distribution layer.
- ③ Access layer.

→ Core Layer

* Core layer Contains Server Room and Server room have Firewall, Router, 2-L3+ Switches, Patch Rack, Server, inverte~~d~~, etc.

→ Distribution layer

* Distribution layer contains 4-Switches.

* The 4 Switches are →

① Main Build → R7 (Room no-213, 2nd Floor).

② Main Build → R1 (Room no-301, 3rd Floor).

③ Main Build → R6 (Room no-113, 1st Floor).

④ Main Build → R2 (Room no-201, 2nd Floor).

* This 4-Switches Contains →

③

① 18 Port Gigabit Easy Smart Switch
With 16 Port. TL-SG1218MPE

② SG3428XMP → 24 Port Gigabit &
4-port 10 G SFP+, L2+ Managed
Switches with 24 Port PoE+.

→ Now Let's discuss these 2 above
Components →

① TP-Link TL-SG1218MPE

* Likely used for : CCTV Camera,
wifi access point, etc.

* 18 Port Gigabit "Easy Smart" Switch.

* 16 Port Support PoE+

② TP-Link Omada SG3428XMP

* Higher-end managed Layer 2+ Switch.

* 24 PoE+ Port, Uplinks, 10 G SFP

* Usually acts as a main distribution
for each floor.

→ Access Layer

* Access layer contain 3-Switches i.e

1) Main Build R1-[Room no-301, 3rd Floor].

2) Main Build R4 [Room no-05, Ground Floor].

3) Main Build R5 [Room no-18, Ground Floor].

* Every switches act as access layer which helps to provide network connectivity to all Classroom, Lab, Faculty Room, etc. ④

* Access Layer directly Connect:

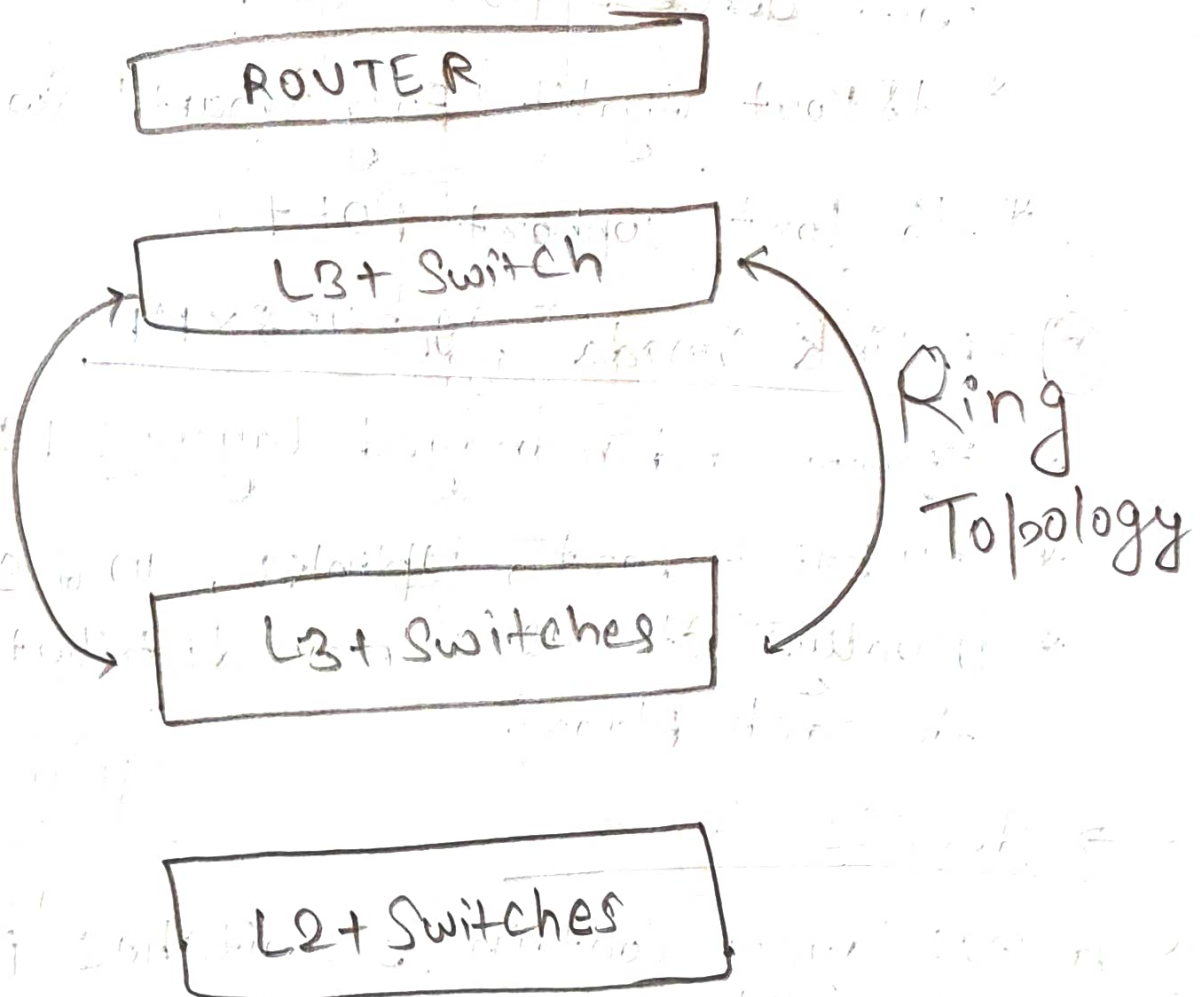
① Ethernet Port

② Wireless AP

③ Lab System

④ outdoor wireless AP (ex-018)

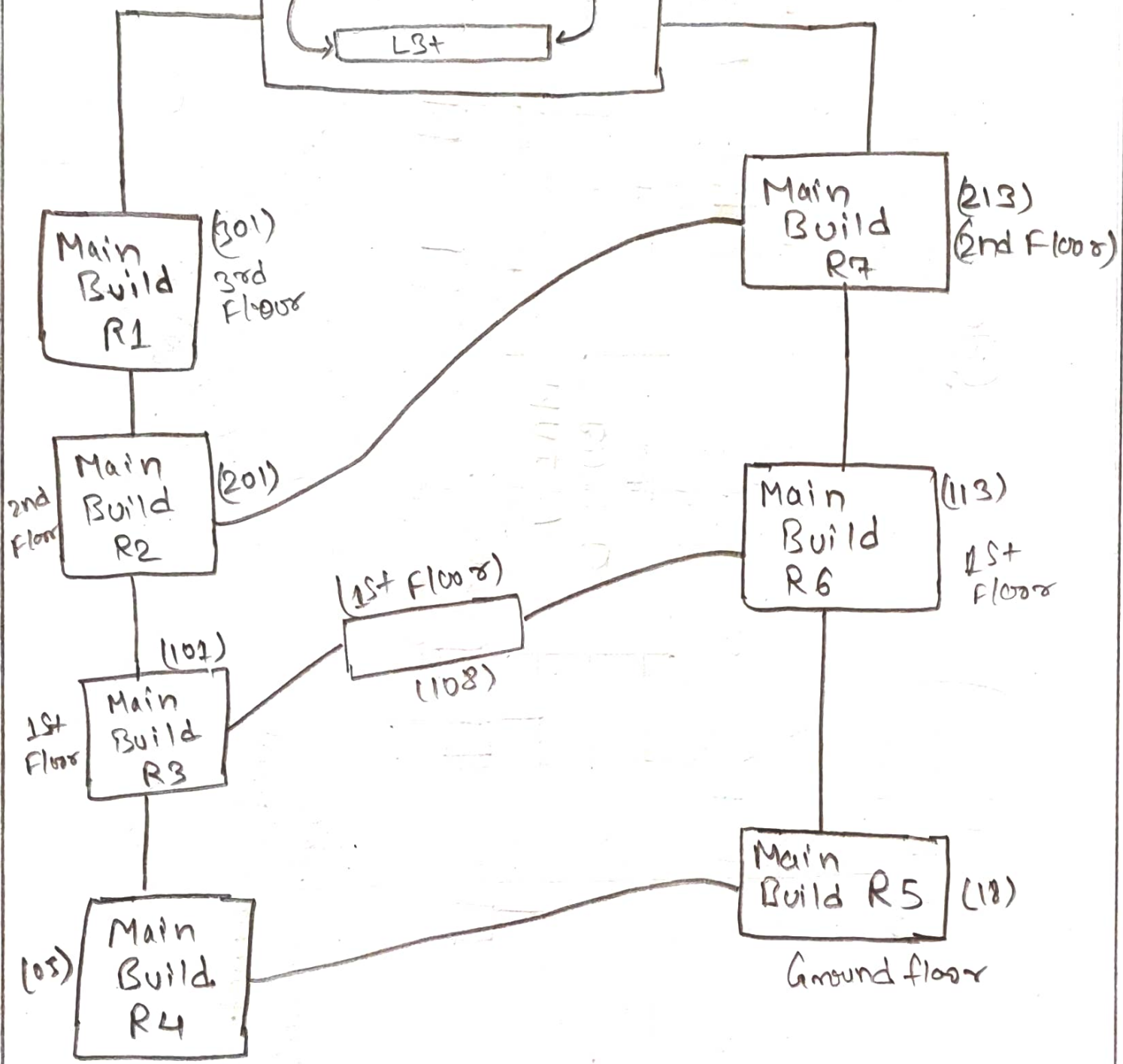
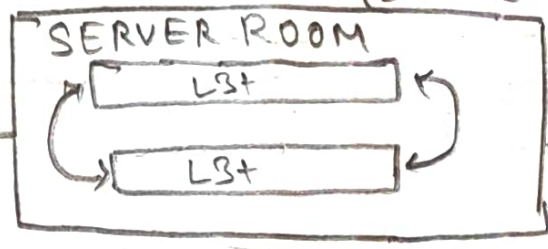
→ SERVER ROOM Architecture/Topology



TOPOLOGY

(210) (2nd Floor)

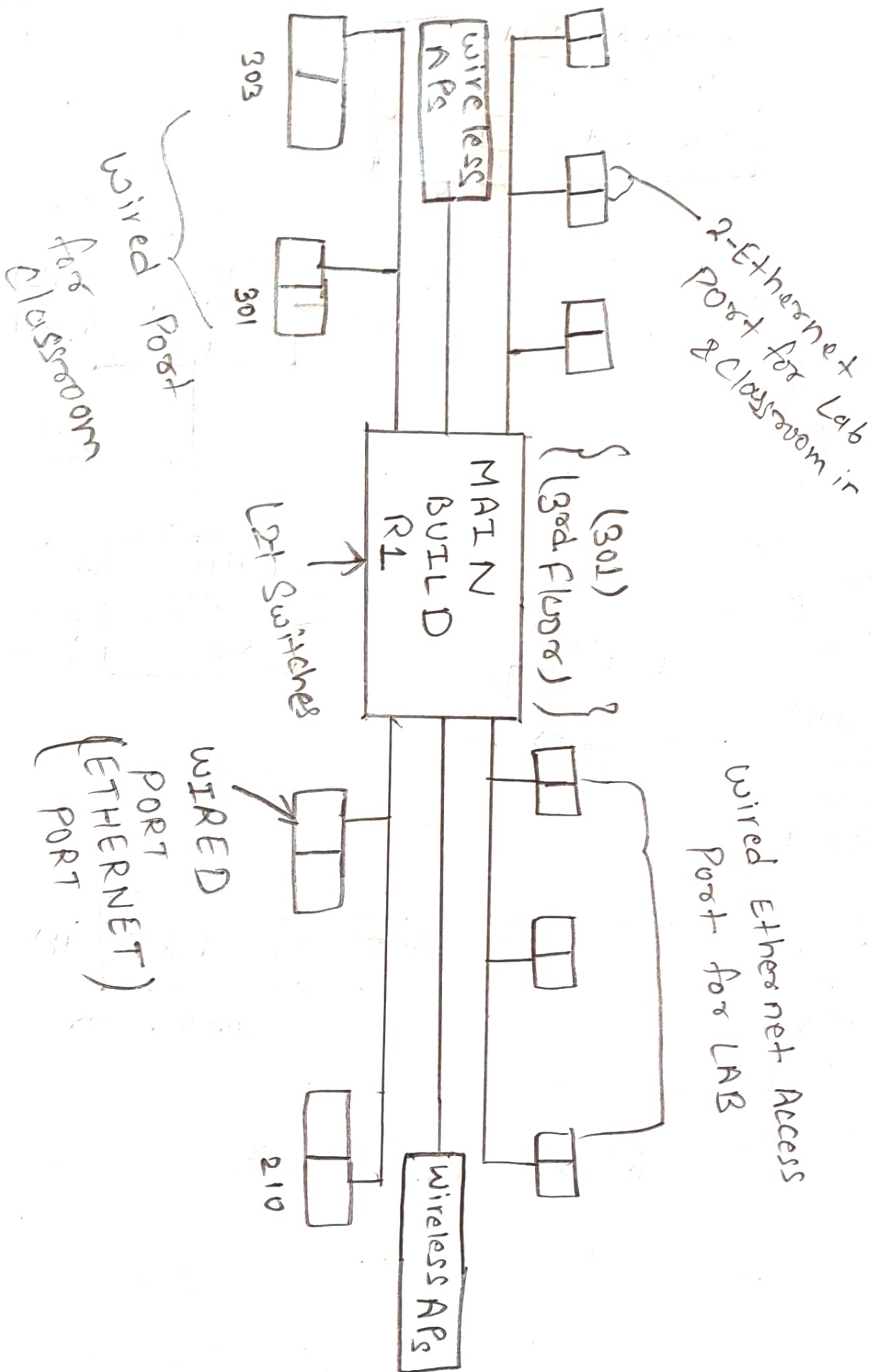
⑤



- R1, R7, R2, R6 → Distribution layer
- Server room → Core layer
- R1, R2, R3, R4, R5, R6, R7 → Access layer

ACCESS LAYER SWITCH (3rd Floor-301)

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WIRELESS CONNECTION

```
C:\Users\conne>ipconfig
```

Windows IP Configuration

Wireless LAN adapter Wi-Fi 2:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi 3:

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::c880:b426:32f0:8dfd%20
IPv4 Address. . . . . : 10.38.232.146
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.38.232.103
```

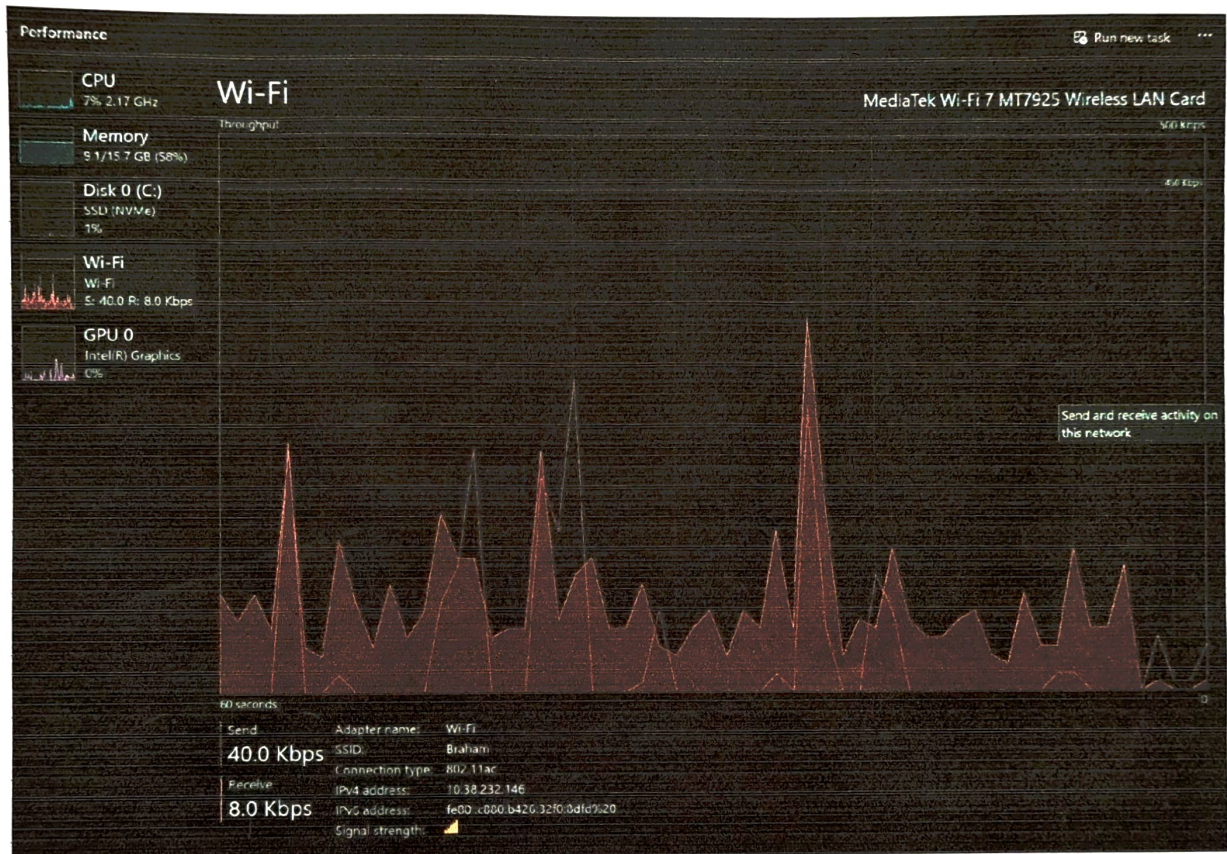
Ethernet adapter Ethernet 3:

```
Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2409:408a:1c8b:f98b:b142:f535:a2d7:cf02
Temporary IPv6 Address. . . . . : 2409:408a:1c8b:f98b:d76:27d0:d0a4:5185
Link-local IPv6 Address . . . . . : fe80::3c29:2fe7:1878:986d%9
IPv4 Address. . . . . : 10.238.72.127
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::d8ed:73ff:fe9f:79de%9
                          10.238.72.213
```

```
C:\Users\conne>
```

↳ using "ipconfig"

WIRELESS SPEED



```
C:\Users\conne>getmac
```

Physical Address	Transport Name
04-68-74-96-8D-8F	\Device\Tcpip_{C6528A0B-6572-4CA0-A806-97A8623E6306}
N/A	Hardware not present

→ MAC ADDRESS
→ using "getmac"

WIRED CONNECTION

```
C:\Users\conne>ipconfig

Windows IP Configuration

Wireless LAN adapter Wi-Fi 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi 3:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix . :
    IPv6 Address. . . . . : 2409:408a:1c8b:f98b:b142:f535:a2d7:cf02
    Temporary IPv6 Address. . . . . : 2409:408a:1c8b:f98b:d76:27d0:d0a4:5185
    Link-local IPv6 Address . . . . . : fe80::3c29:2fe7:1878:986d%9
    IPv4 Address. . . . . : 10.238.72.127
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::d8ed:73ff:fe9f:79de%9
                                10.238.72.213

C:\Users\conne>
```

using IPCONFIG

→ Difference b/w wired & wireless speed?

Soln

Wired Network

* Faster Communication

* Stable Connection

* Low latency

* Dedicated physical medium

Wireless Network

* Slower Communication.

* Affected by distance / Interference

* Higher latency

* Shared radio medium.

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

Q Role of Subnet mask in network?

→ Ans

A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

* It is important for:

- ① Network Segment
- ② Efficient routing
- ③ Communication b/w devices.

→ OBSERVATION

During the study of the College network infrastructure :-

- ① A Centralized Server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-Link PoE+ Switches were used in network cabinets.

→ Advantage of Existing Network?

- * Scalable architecture.
- * Centralized management.
- * Supports PoE+ device.
- * Easy expansion of floors & Classroom.
- * Support both wireless & wired.
- * Redundant link for failover.

CONCLUSION

The observed campus LAN follow a hybrid hierarchical topology consist of tree, Extended star, Ring. The network uses managed PoE Switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.